

# AAMIR KHAN

## FIRST-CLASS COMPUTER SCIENCE GRADUATE | SOFTWARE ENGINEERING

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### PROFILE

First-Class Computer Science graduate with project experience building Python APIs, authenticated web applications, relational databases and server-authoritative multiplayer systems. Strong academic results across software engineering, web development and algorithms.

### EDUCATION

**University of Leeds** | *BSc Computer Science, First-Class Honours*

**Sep 2023 - Jul 2026**

- Overall average 72.4%; dissertation: Algorithmic Game Playing and Networked Implementation for the Nonaga Strategy Game - 70% (40 credits).
- Selected results: Web Application Development 96%, Object-Oriented Programming 86%, Machine Learning 80%, Software Engineering Principles 80%.
- International Excellence Scholarship recipient; received £5,000 across the degree, with continuation awards based on academic performance.

### TECHNICAL SKILLS

**Languages:** Python, JavaScript, SQL, C++, Java

**Backend & Web:** FastAPI, Flask, SQLAlchemy, REST APIs, Jinja2, Flask-SocketIO, HTML, CSS, Bootstrap

**Databases:** PostgreSQL, MySQL, SQLite, pgvector

**Testing & Tools:** pytest, Git, Docker, GitHub Actions, Alembic, Render, PythonAnywhere

### SELECTED PROJECTS

**ScholarGraph** | *Individual*

**Jan 2026**

- Built and deployed an asynchronous FastAPI research platform with 29 endpoints, integrating arXiv metadata, Semantic Scholar citation graphs and 384-dimensional paper embeddings stored in PostgreSQL with pgvector.
- Implemented JWT authentication, Alembic migrations, scheduled crawling with retries and exponential backoff, Docker, GitHub Actions and MCP access for AI clients; created 32 tests with 59% coverage.

**Nonaga Strategy Game** | *Individual dissertation*

**Apr 2026**

- Designed a fixed-depth minimax AI with alpha-beta pruning and three handcrafted heuristics, balancing search quality against latency: approximately 0.9 seconds at depth 3 versus 8 seconds at depth 4.
- Built a server-authoritative Flask-SocketIO multiplayer system with room-based state, action validation, reconnect tokens and a 60-second recovery window; created 54 passing pytest tests across eight files.

**RepIT Fitness Tracker** | *Individual*

**Mar 2025**

- Built a Flask fitness platform supporting saved routines, live workout sessions, set/repetition/weight logging and body-measurement charts.
- Designed a relational SQLAlchemy schema and integrated 1,500+ locally cached ExerciseDB exercises; implemented Flask-Login authentication, password hashing and protected user-specific routes.

**Auction Website** | *Team of 6*

**May 2025**

- Delivered 17 backlog features - 14 independently and three through pair programming - across listings, search, role-based dashboards, expert authentication workflows and responsive UI.
- Integrated Stripe test payments, email notifications, live countdowns and Socket.IO auction updates, with server-side bid validation and transactional SQLAlchemy updates.

### EXPERIENCE

**BISAG-N, New Delhi** | *Python Developer Intern*

**Jun - Aug 2024**

- Shadowed engineers in a professional technology environment and strengthened Python skills through structured independent practice.
- Served as a rapporteur at the Global India AI Summit, producing a concise internal summary of talks and discussions from senior technology leaders.

**Avisha Association, Noida** | *Technical Volunteer*

**Jul - Sep 2023**

- Independently built a Python/MySQL GUI application for creating, editing, deleting and searching water-pot records used by the NGO owner in the local office.